

CS & IT ENGINEERING

Algorithms

Greedy Method

Lecture No.- 05



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

OMP

Huffman Encoding

Topics to be Covered



Topic

Topic

Topic

Huffman Encoding

MCST



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

TTTe



Topic : Greedy Algorithms

$A \rightarrow 0.23 \dots\dots\dots(4)$

$B \rightarrow 0.05 \dots\dots\dots(1)$

$C \rightarrow 0.48 \dots\dots\dots(5)$

$D \rightarrow 0.15 \dots\dots\dots(3)$

$E \rightarrow 0.09 \dots\dots\dots(2)$

Priority (low to high)





Topic : Greedy Algorithms

Eg..

Given: Language/ Elements $\Rightarrow \{A,B,C,D,E\} \rightarrow 5$ character

If uniform encoding: $\lceil \log_2 N \rceil$ bits character

$$= \lceil \log_2 5 \rceil$$

$$= \underline{\underline{3 \text{ bit / character}}}$$

Note:- Start from lowest level.



Topic : Greedy Algorithms

2-way Optimal Binary merge tree:

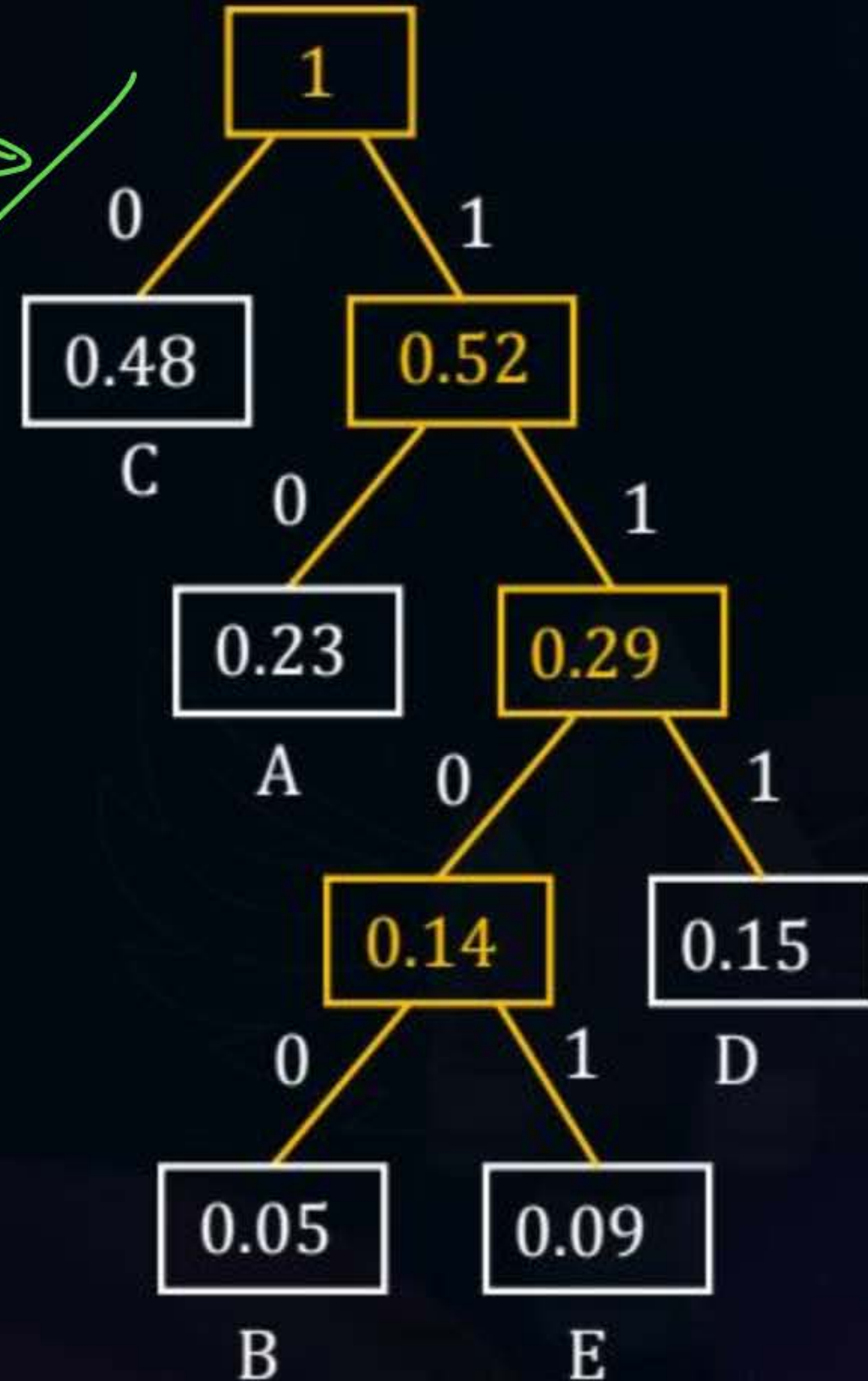
Left Branch $\rightarrow 0$

Right Branch $\rightarrow 1$

Standard

Left $\rightarrow$ Smaller

Right $\rightarrow$ Larger





Topic : Greedy Algorithms

A $\rightarrow$ 10 $\rightarrow$ 2 bits

B $\rightarrow$ 1100 $\rightarrow$ 4 bits

C $\rightarrow$ 0 $\rightarrow$ 1 bits

D $\rightarrow$ 111 $\rightarrow$ 3 bits

E $\rightarrow$ 1101 $\rightarrow$ 4 bits

Non-Uniform encoding & all are
unique



Topic : Greedy Algorithms

#Q. ~~Average~~ number of bit required using Huffman encoding? Coding:

Arg

= Weighted external path length:

$$= \sum_{i=1}^n d_i * q_i$$

Hw



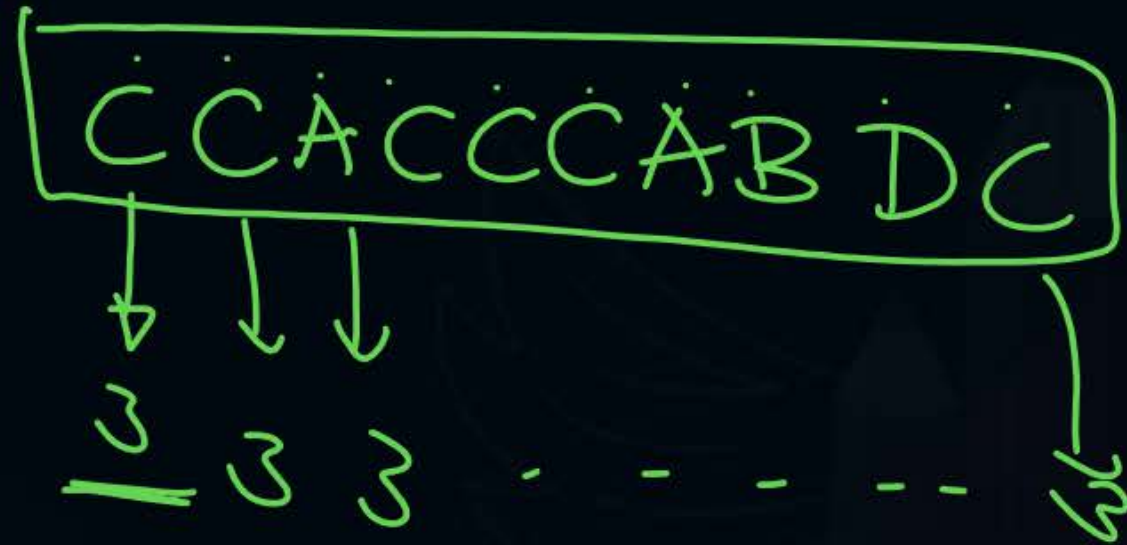
Topic : Greedy Algorithms

#Q. "CCACCCA BDC"

How many bit required to above text using:-

1. Uniform Encoding: $= 3 \times 10 = 30 \text{ bits}$

2. Huffman Encoding:
(Non-uniform)





Topic : Greedy Algorithms

2. Using Huffman Encoding:

Text: C C A C C C A B D C

→ [0 0 10 0 0 0 10 1100 111 0]

Final Encoding:

[0 0 1 0 0 0 0 1 0 1 1 0 0 1 1 1 0]

Total = 17 bit compressions

<u>Appx 2 :-</u>	<u>For</u>
A $\rightarrow$ 2 bits	$\rightarrow$ 2
B $\rightarrow$ 4 bits	$\rightarrow$ 1
C $\rightarrow$ 1 bit	$\rightarrow$ 6
D $\rightarrow$ 3 bits	$\rightarrow$ 1
E $\rightarrow$ 4 bits	$\rightarrow$ 0

$$\begin{aligned}
 &\text{Total bits} \\
 &= 2 \times 2 + 1 \times 4 + 6 \times 1 + 1 \times 3 + 0 \times 4 \\
 &= \textcircled{17}
 \end{aligned}$$



Topic : Greedy Algorithms

#Q. The characters 'A' to 'H' have the set of frequencies based on the first 8 Fibonacci number as follows:

A = 1, B = 1, C = 2, D = 3, E = 5, F = 8, G = 13, H = 21

Types A Huffman code is used to represent the characters. What is the sequence of characters corresponding to the following code: 1101111001¹010_^

A

FDHEG ✓

C

DCHFG ✗

B

ECGDF ✗

D

FEHDG ✗



Topic : Greedy Algorithms

A → 1111100

B → 1111101

C → 111111

D → 11110

E → 1110

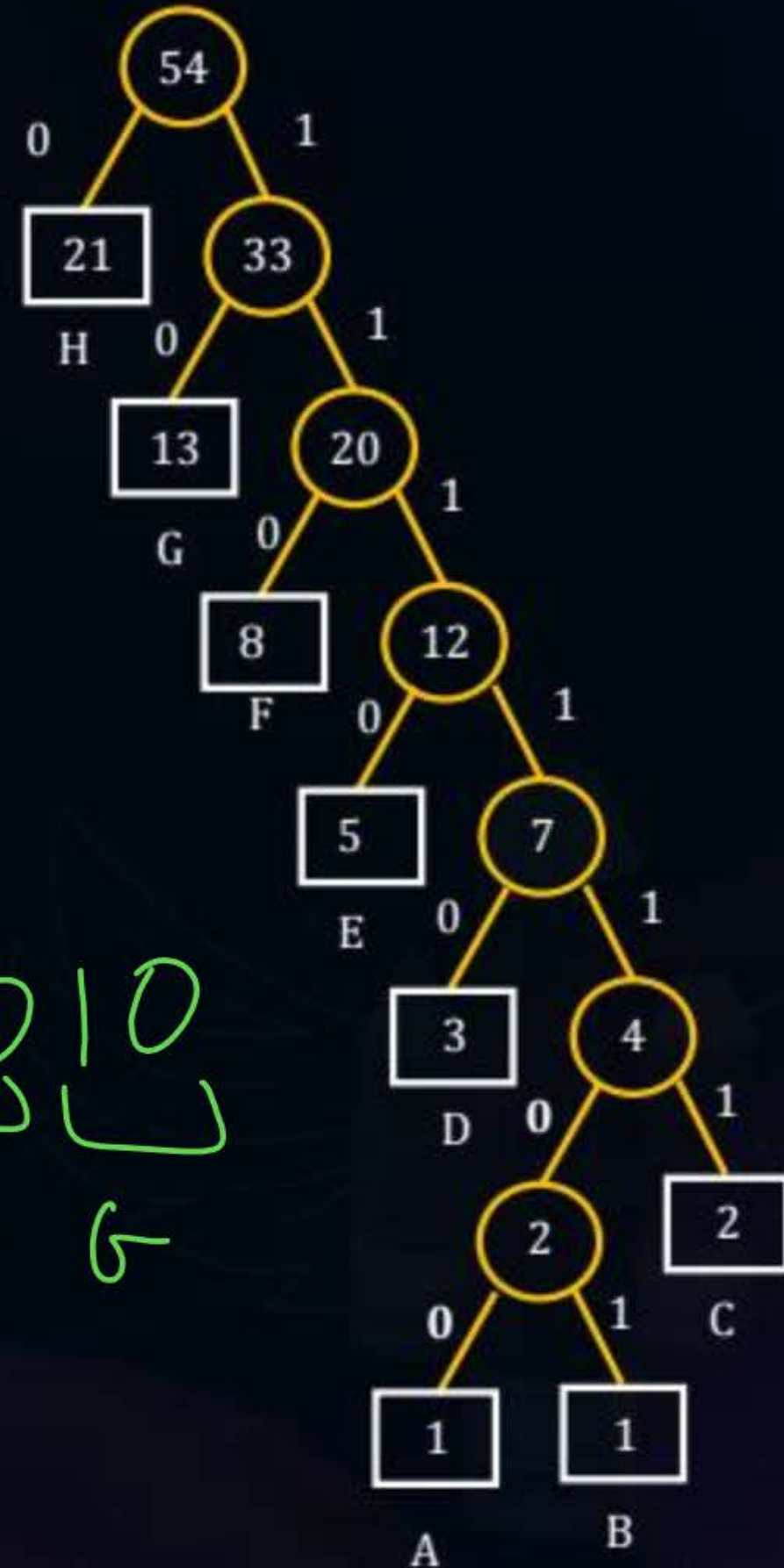
F → 110

G → 10

H → 0

Encoding

110111100111010
F D H E G





Topic : Greedy Algorithms

<u>110</u>	<u>11</u> <u>110</u>	<u>0</u>	<u>111</u>	<u>010</u>
F	D	H	E	G

Roof $\rightarrow$ leaf

“FDHEG”



Topic : Greedy Algorithms

19✓

#Q. A message is made up entirely of characters from the set $X = \{P, Q, R, S, T\}$. The table probabilities for each of the characters is shown below:

Characters	Probability
P	0.22 ✓
Q	0.34
R	0.17 ✓
S	0.19 ✓
T	0.08 ✓
Total	1.00

Type 6

0.25, 0.34, 0.41

If a message of 100 characters over X is encoded using Huffman coding, then the expected length of the encoded message in bits is_____.



Topic : Greedy Algorithms

P → 01

Q → 11

R → 101

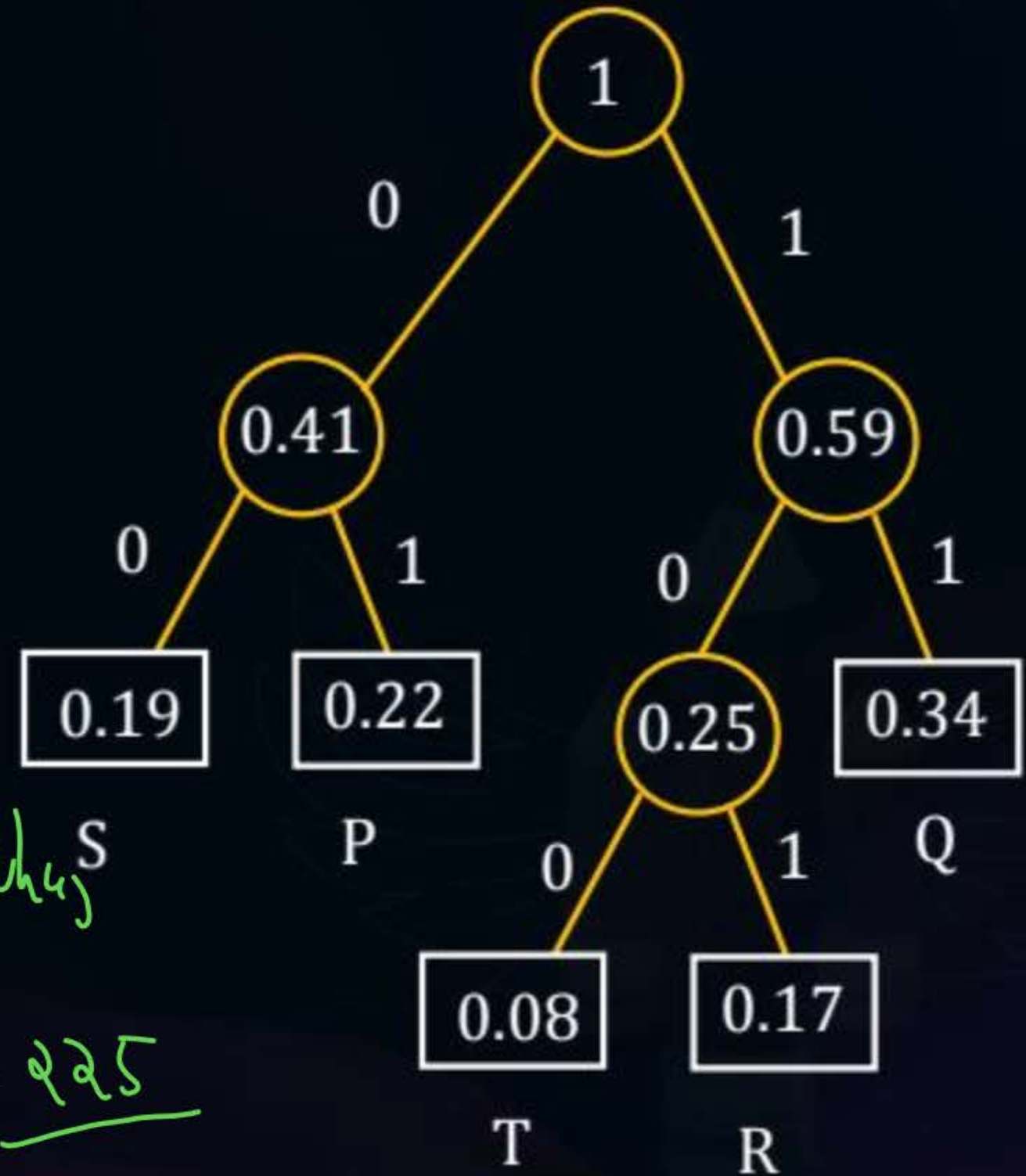
S → 00

T → 100

$$\begin{array}{r} 1 \\ 0.41 \\ 0.59 \\ 0.25 \\ \hline 2.25 \\ \hline \text{HW} \end{array}$$

Avg bits/char = 2.25 bits/char

$$100 \text{ char} = 2.25 \times 100 = \underline{225}$$



[MCQ]

#Q. Consider the string abbccddeee. Each letter in the string must be assigned a binary code satisfying the following properties:

- Type 7
1. For any two letters, the code assigned to one letter must not be a prefix of the code assigned to the other letter.
 2. For any two letters of the same frequency, the letter which occurs earlier in the dictionary order is assigned a code whose length is at most the length of the code assigned to each other letter.

Among the set of all binary code assignment which satisfy the above two properties, what is the minimum length of the encoded string?

A 25

B 23

C 21

D 30



Topic : Greedy Algorithm

Encoding:

a → 100 (3)

b → 00 (2)

c → 01 (2)

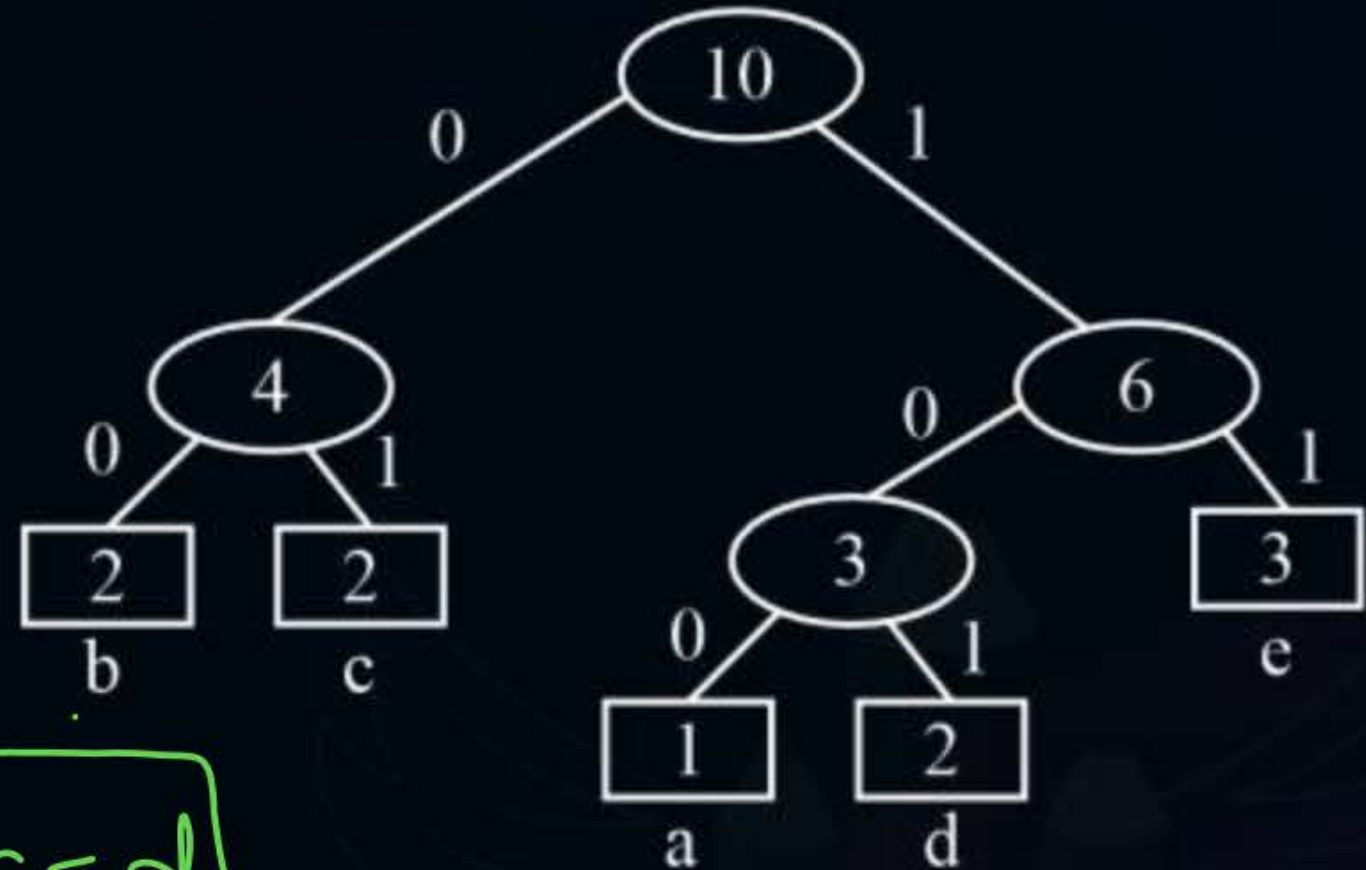
d → 101 (3)

e → 11 (2)

Text "abbccddeee"

$$\begin{aligned}\text{Total bits} &= 3 \times 1 + 2 \times 2 + 2 \times 2 + 3 \times 2 + 2 \times 3 \\ &= 23\end{aligned}$$

Handwritten notes:
a → 1 — 1)
b → 2 — 4)
c → 2 — 3)
d → 2 — 2)
e → 3 — 5)
b ≤ c ≤ d





Topic : Greedy Algorithm



Minimum Cost Spanning Tree (MCST):

- Graph Based Problem
- Graph : $G \equiv (V, E)$
where, $V =$ Set of vertices
 $E =$ Set of edges



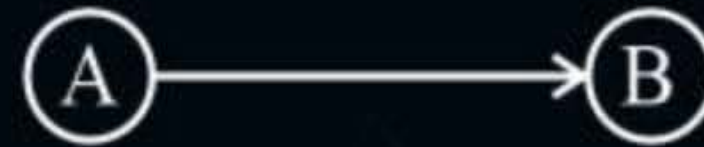
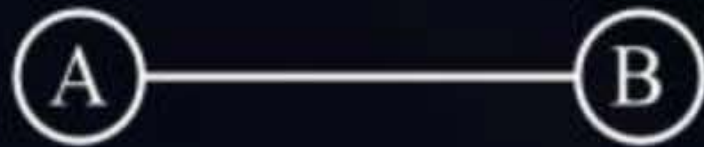


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Types:

(1) Undirected vs Directed:



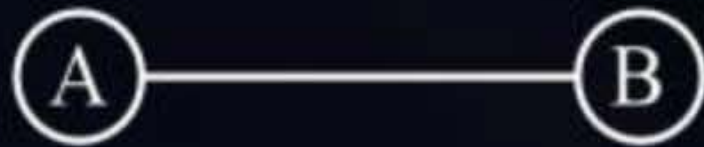


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Types:

(2) Weighted vs Un-weighted:

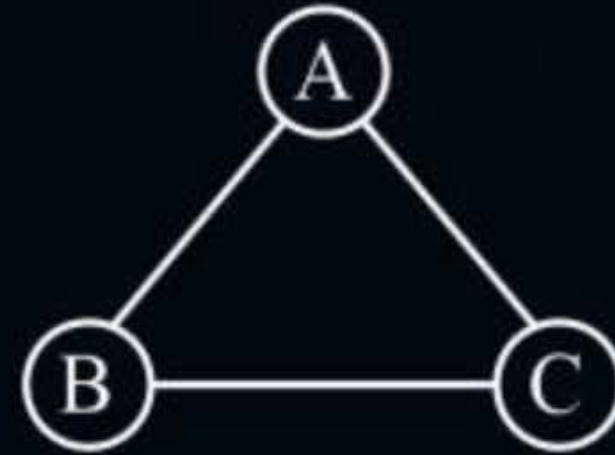




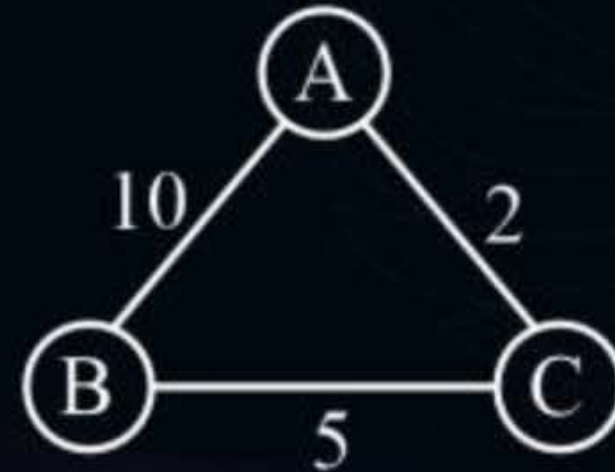
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(1) Undirected Unweight Graph:



(2) Undirected Weight Graph:

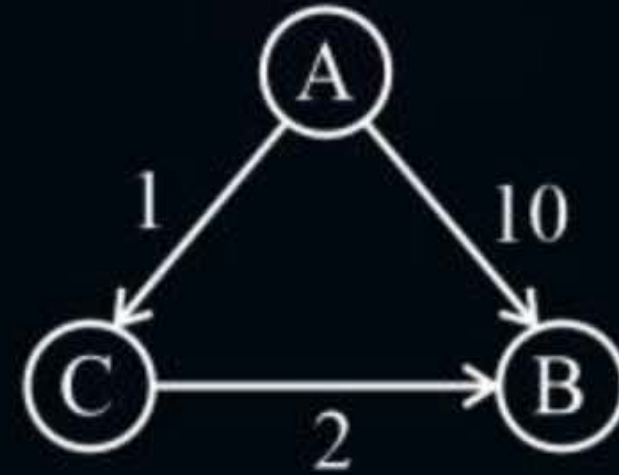




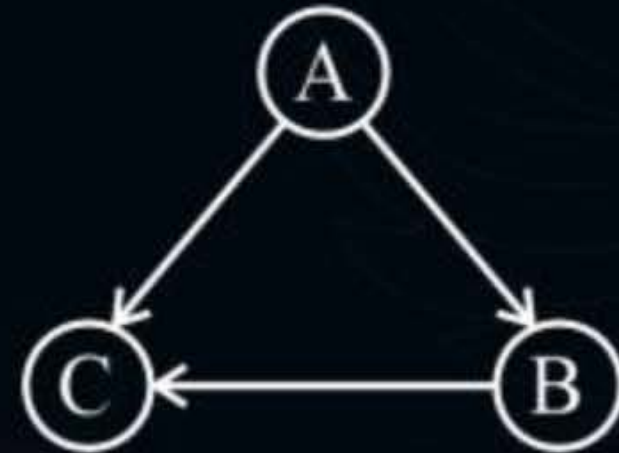
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(3) Directed weight Graph:



(4) Directed Unweight Graph:





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Representation of Graph:

- (1) Adjacency matrix → (Array) ✓
- (2) Adjacency List → (List based) ✓



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(1) Adjacency matrix:

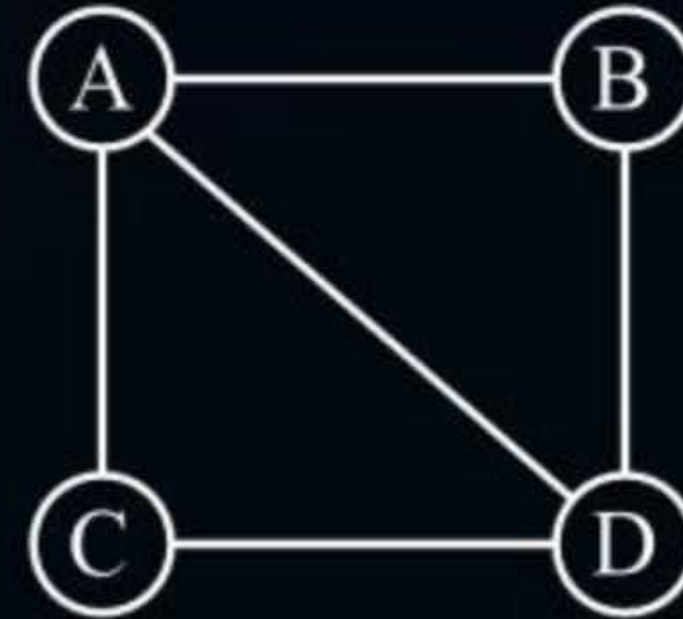
$n = \text{no. of vertices} = 4$

$e = \text{no. of edges} = 5$

Size = $O(n^2)$

$\text{Adj}[i, j] = \underline{1}$ $(i, j) \in E$

$\text{Adj}[i, j] = \underline{0}$ $(i, j) \notin E$



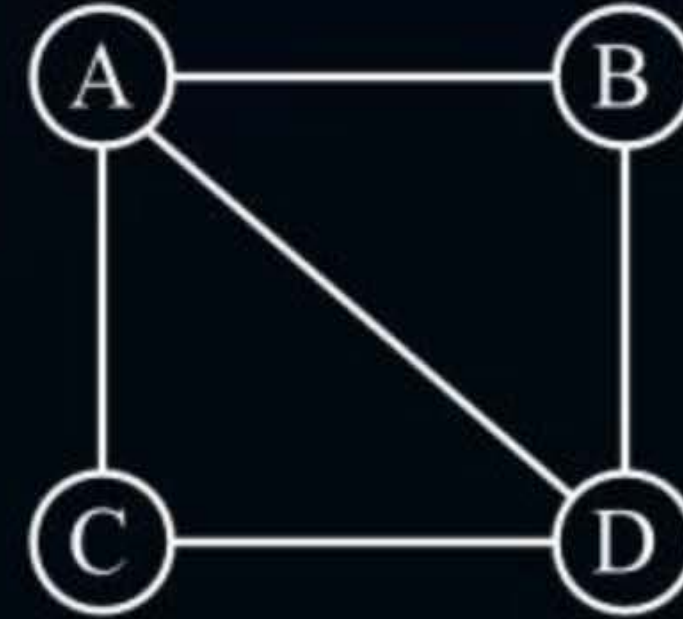
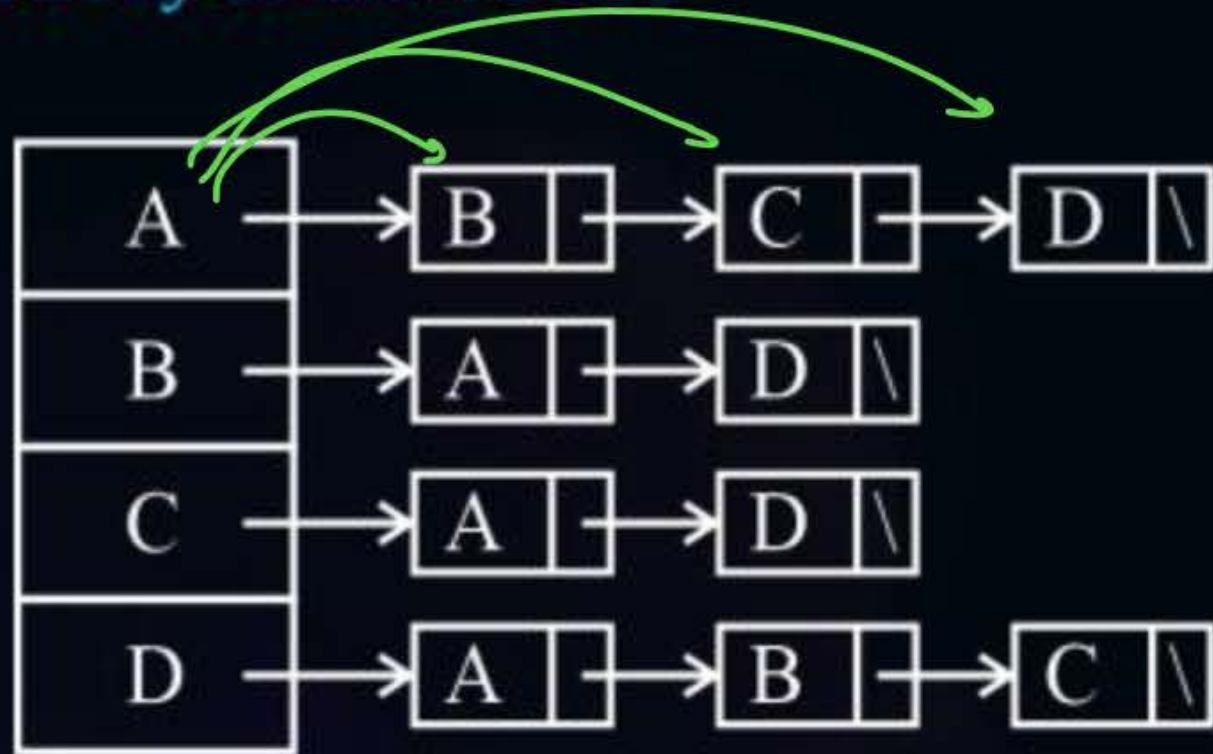
Adj	A	B	C	D
A	0	1	1	1
B	1	0	0	1
C	1	0	0	1
D	1	1	1	0



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(2) Adjacency List:

Array of linked list



(i) Undirected graph: $(n + 2e)$

(ii) Directed graph: $(n + e)$

$\} \rightarrow O(n + e)$



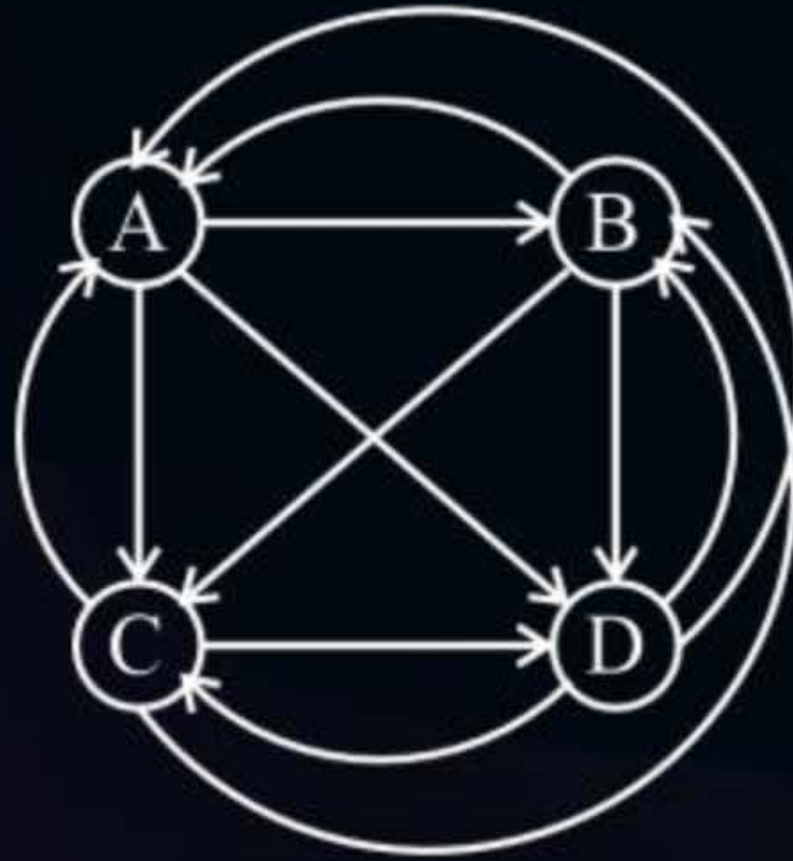
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(1) Directed Complete Graph:

$$e = n * (n - 1)$$

$$\rightarrow \underline{O(n^2)}$$





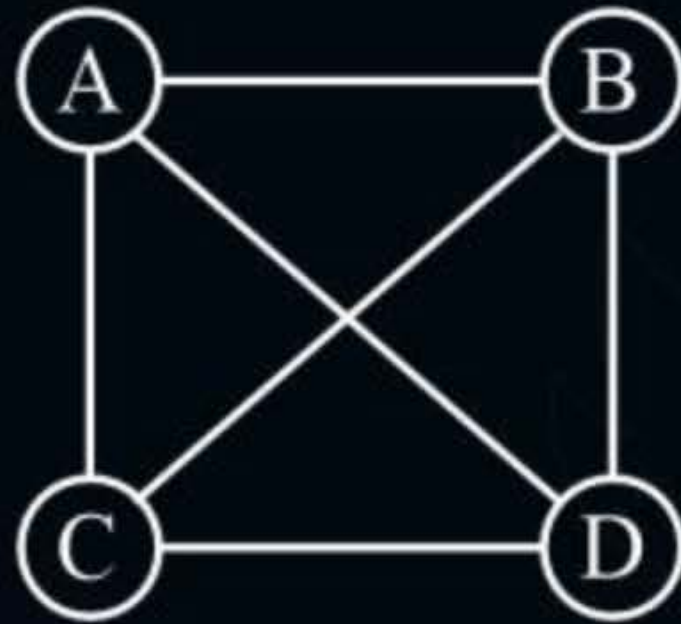
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(2) Un-Directed Complete Graph:

$$e = n(n - 1) / 2$$

$$\rightarrow \underline{\underline{O(n^2)}}$$



[NAT]

#Q. Which representation to use when?

Adj Matrix $\rightarrow$ $O(n^2)$

Adj List $\rightarrow$ $O(n + e)$



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Dense Graph:

Complete Graph:

- $e \approx \underline{O(n^2)}$
- Matrix refer is preferred.

Sparse Graph:

Non-Complete Graph:

$$e \neq O(n^2)$$

- Adj list is preferred.
- $O(n + e)$



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Spanning Tree:

- A subgraph $T(V, E')$ of a given graph (V, E) where $E' \subset E$ is a spanning Tree, iff T is a Tree.



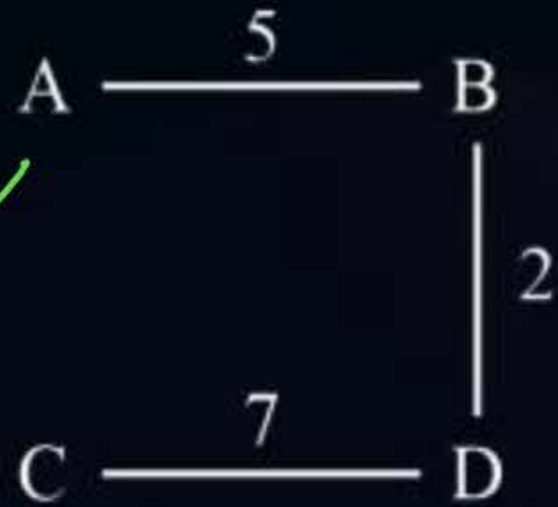


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Example:

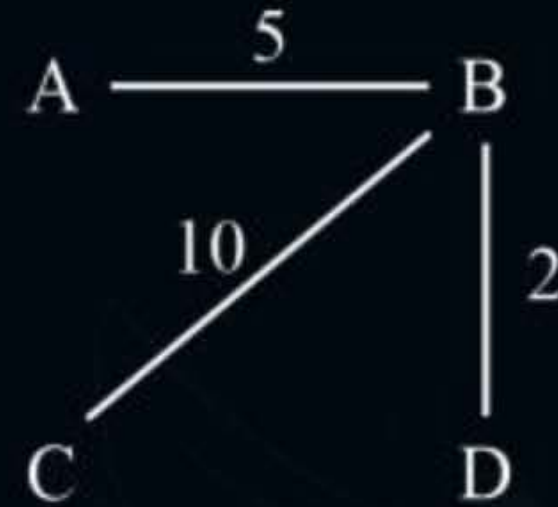
Q. Which of the following are valid Spanning Tree of G?

A

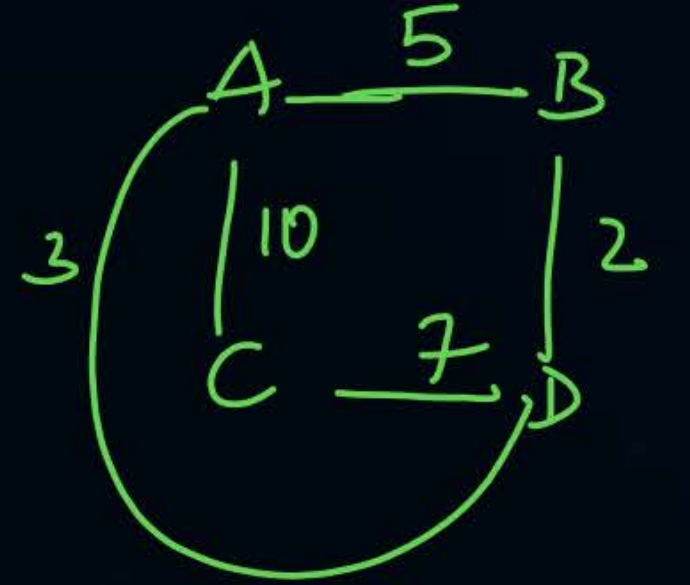
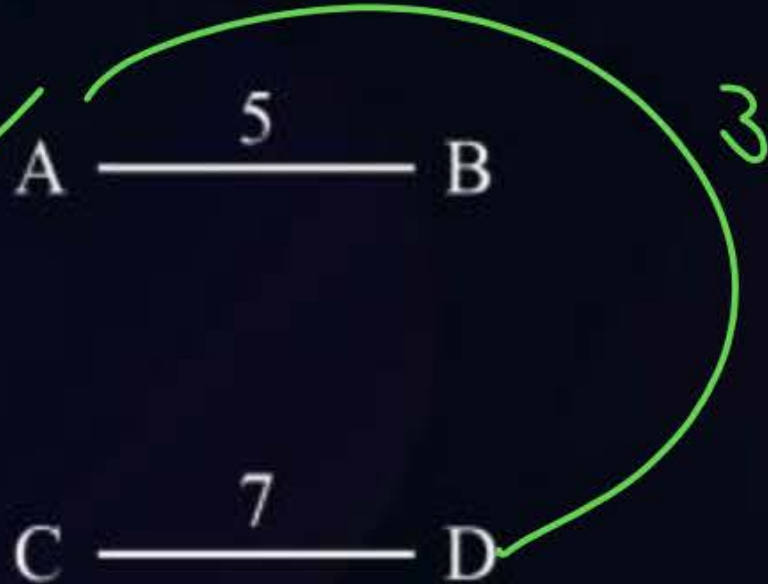


A & C

B



C





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Important Observations:

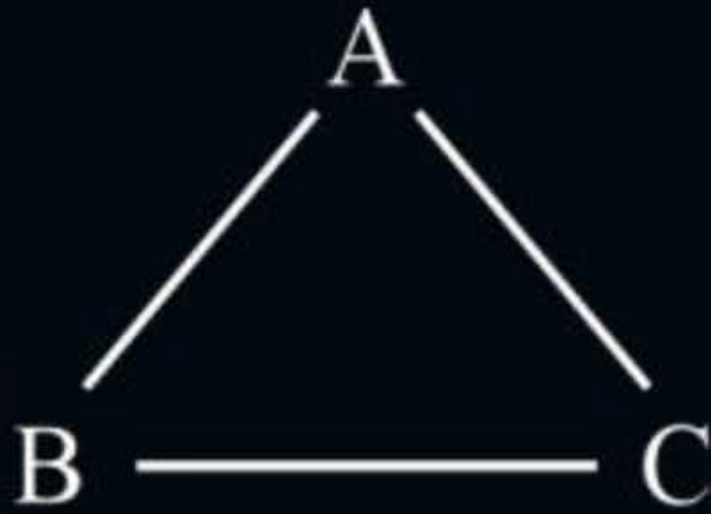
- 1) Multiple Spanning Trees are possible

$$ST - 1 \rightarrow 5 + 2 + 7 = 14$$

$$ST - \underline{3} \rightarrow 5 + \underline{2} + 7 = \underline{14} / \underline{15}$$

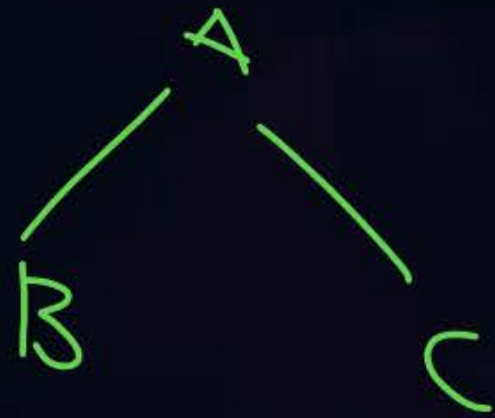
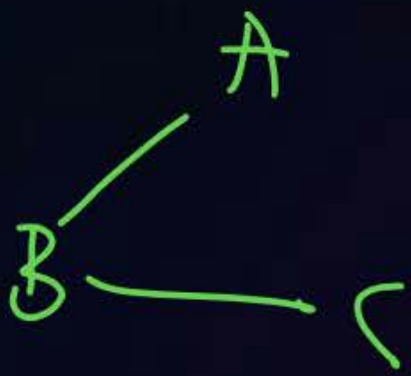
- 2) A Spanning Tree with ~~n-matrices~~ vertices will always have (n - 1) edges.

#Q. For a given graph with n -vertices and e -edges the maximum number of Spanning Tree =?



$$\approx n^{(n-2)}$$

$$\Rightarrow \approx \underline{\underline{O(n^n)}}$$



[NAT]

#Q. Graph: $n \rightarrow$ vertices,
 $e \rightarrow$ edges

The no. of edges that must be removed from this graph to get its Spanning Tree?

(8)



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- Spanning Tree of 'n' vertices → (n - 1) edges required.

Total edges = (required + removal)

↓ ↓ ↓

$$e = (n-1) + x$$

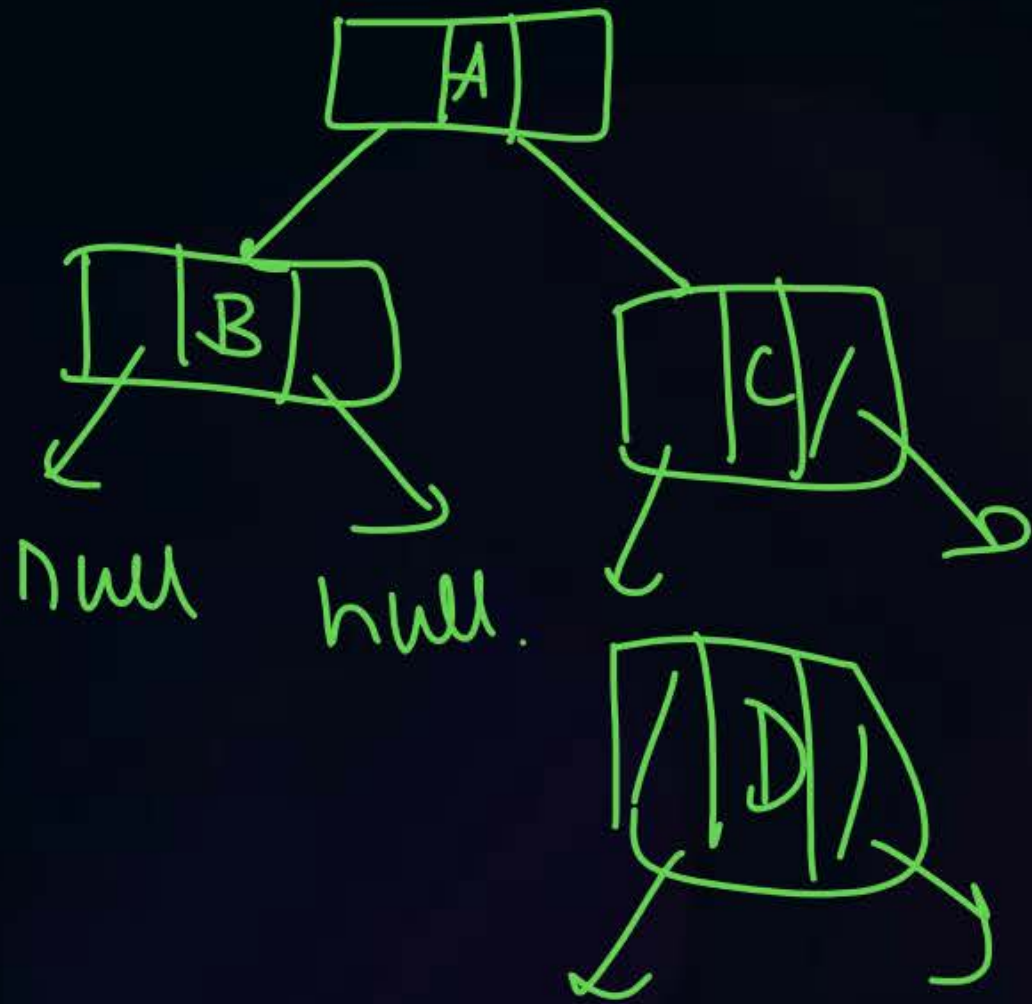
$$x = e - (n - 1)$$

$$x = e - n + 1$$

[NAT]

#Q. A Tree with 'n' nodes/vertices will have _____ no. of null link?

eg:-





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Logic: Total links in a tree with n vertices? = $2 * n$

edges (used) = $(n - 1) \rightarrow$ non-null links

unused/null-links = Total - edges

$$= 2n - (n - 1)$$

$$= 2n - n + 1$$

$$= n + 1$$





2 mins Summary



Topic

Topic

Topic

Topic

Topic

Huffman Encoding
Intro to MCST.



THANK - YOU